Automatic Identification of Slavic Languages from Audio Using Deep Learning

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Abstract

Spoken language identification (LID) for closely related languages is hard: shared phonology, lexical overlap, and short clip durations all work against the classifier. We study this problem for eight Slavic languages spanning all three branches of the family – East (Belarusian, Russian, Ukrainian), West (Czech, Polish), and South (Bulgarian, Macedonian, Serbian) – using crowd-sourced speech from Mozilla Common Voice as the primary training and in-domain test source, and studio-recorded Google FLEURS as an out-of-domain benchmark. We design and compare three tiers of deep learning systems of increasing capacity, from shallow CNNs trained from scratch to fine-tuned large pre-trained speech transformers, varying input features, data augmentation, and training objectives along the way. Our best system, a fine-tuned Whisper-small encoder with a two-layer classification head, achieves a macro-F1 of **0.795** on the in-domain test set and **0.817** on FLEURS, demonstrating strong cross-domain generalization. Three findings stand out: (1) the pre-training task is the dominant factor – a smaller model with task-aligned pre-training outperforms a larger one without it; (2) Belarusian, Russian, and Ukrainian form a persistent confusion cluster at every tier; and (3) Czech shows severe acoustic domain shift between crowdsourced and read speech, collapsing under weaker models while recovering well under Whisper.

1 Introduction

While Slavic languages exhibit high degrees of mutual intelligibility for native speakers, automatic spoken language identification (LID) within this family remains highly challenging due to severe acoustic and lexical overlap. As a foundational task in multilingual speech processing pipelines, accurate LID enables downstream systems such as automatic speech recognition and machine translation to select the appropriate language-specific model, and is therefore a prerequisite for robust multilingual voice interfaces. While LID has been studied extensively for major world languages [14, 9], the Slavic language family presents a particularly challenging setting: the eight languages considered here span three distinct branches, yet share deep lexical and phonological similarities within each branch that can confuse acoustic classifiers.

This project addresses 8-way Slavic LID from 3-second audio clips. Our primary contributions are:

1. A comprehensive empirical study of three model tiers – shallow CNN, intermediate acoustic encoders, and fine-tuned pre-trained transformers – applied to the same eight-language task, enabling clean tier-to-tier comparisons.
2. A systematic investigation of how input features (log-Mel vs. MFCC with deltas), data augmentation (SpecAugment, Mixup), and training objectives (cross-entropy, label smoothing, AAM-Softmax) affect performance at each tier.
3. Evaluation on both in-domain (MCV) and out-of-domain (FLEURS) benchmarks, with per-language analysis revealing Czech domain collapse, the Polish out-of-domain advantage, and persistent East Slavic confusion.
4. A finding that supervised multilingual pre-training (Whisper-small, 87M parameters) substantially outperforms larger self-supervised models (XLS-R-300M, 300M+ parameters) on this task, macro-F1 0.795 vs. 0.613.
5. A supplementary dialect experiment showing that the same Whisper backbone, adapted on the BDLT corpus, achieves macro-F1 0.867 on 4-way Bulgarian regional dialect identification with only ∼200 source recordings.

2 Related Work

Language identification from speech. Traditional LID systems relied on i-vector or x-vector embeddings extracted from MFCCs [14], often combined with language model back-ends. More recent approaches apply end-to-end deep learning directly on spectrograms or waveforms [5, 13].

Slavic speech resources. Mozilla Common Voice (MCV) [2] provides crowdsourced recordings for all eight languages studied here. VoxLingua107 [15] is a purpose-built LID dataset covering 107 languages including Slavic ones. Google FLEURS [6] provides read-speech recordings useful as an out-of-domain evaluation benchmark.

Pre-trained speech models. XLS-R [3] extends wav2vec 2.0 [4] to multilingual self-supervised pre-training on 436 languages. Whisper [12] is trained with weakly supervised multilingual ASR on 680k hours of labelled audio, covering numerous Slavic languages.

Metric learning for speaker and language recognition. AAM-Softmax (ArcFace) [7], originally proposed for face recognition, has become the standard training objective for speaker verification (x-vectors [14], ECAPA-TDNN [8]) due to its ability to learn compact, discriminative embeddings. We investigate whether the same benefit transfers to language identification.

3 Dataset

3.1 Sources and Splits

We use a composite dataset drawn from three publicly available sources:

- **Mozilla Common Voice (MCV)**: crowd-sourced read speech, primary source for both training and in-domain testing.
- **VoxLingua107**: YouTube-derived speech segments, used to supplement the training set.
- **Google FLEURS**: studio-quality read speech, used exclusively as an out-of-domain evaluation benchmark.

Table 1: Dataset split statistics.

Split	Samples	Source
Train	80 000	MCV + VoxLingua
Test	107 439	MCV only
FLEURS (OOD)	6 157	FLEURS only

All audio is resampled to 16 kHz mono. Each clip is either truncated or zero-padded to exactly 3 seconds (48 000 samples). The eight target languages are listed in Table 2.

Table 2: Target languages

Code	Language	Branch	Test samples
be	Belarusian	East	29 232
ru	Russian	East	19 069
uk	Ukrainian	East	16 501
cs	Czech	West	14 441
pl	Polish	West	16 949
bg	Bulgarian	South	6 318
mk	Macedonian	South	3 263
sr	Serbian	South	1 666

3.2 Class Imbalance

The test set exhibits a 17.5× imbalance ratio between the largest class (Belarusian, 29 232 samples) and the smallest (Serbian, 1 666 samples). This makes micro-accuracy a misleading headline metric, as a model predicting only Belarusian would achieve 27.2% micro-accuracy. We therefore use **macro-F1** (unweighted mean of per-class F1 scores) as our primary evaluation metric throughout. Micro-accuracy is reported for completeness.

3.3 Out-of-Domain Evaluation

The FLEURS split serves as our out-of-domain benchmark. Unlike MCV, FLEURS recordings are studio-quality read speech, introducing an acoustic domain shift. Crucially, FLEURS is approximately class-balanced, which means FLEURS macro-F1 and test macro-F1 are not directly comparable: apparent "OOD gains" can arise simply from evaluating on a balanced set. We therefore compare per-language scores individually when assessing OOD generalisation.

4 Model Architectures

We evaluate three tiers of model capacity. Within each tier we train several configurations varying the input features, augmentation strategy, and training objective.

4.1 Tier 1, Shallow CNN

The baseline model is a four-block convolutional neural network. Each block applies Conv2D (3×3, padding=1) → BatchNorm → ReLU → MaxPool2D(2,2), with channel widths 1→32→64→128→256. The final feature map is reduced to a 256-dimensional embedding via global average pooling, followed by a linear classifier.

Input variants.

- *Log-Mel spectrogram*: 80 mel bins, 25 ms Hann window, 10 ms hop → shape (80, 300). Per-sample instance normalisation applied.
- *MFCC*: 40 coefficients per frame with first and second derivatives concatenated → 120-dimensional feature per frame, same window and hop as above.

Augmentation. SpecAugment [11] is applied during training: two frequency masks (max 15 bins) and two time masks (max 30 frames). For one configuration, Mixup ($\alpha = 0.2$) is additionally applied.

Training objective variants. Cross-entropy was used with ($\varepsilon = 0.1$) or without label smoothing.

4.2 Tier 2, Intermediate Acoustic Encoders

CNN + BiLSTM. Four CNN blocks (same as Tier 1) reduce the frequency axis, outputting a sequence of feature vectors. A 2-layer bidirectional LSTM (hidden 256, total 3.0M parameters) processes the time dimension, followed by attentive statistics pooling [10] to produce a fixed-length embedding.

CNN + Transformer encoder. Two CNN blocks pool along the frequency axis only (kernel (2,1)), mapping log-Mel (64, 301) to a feature map of shape (128, 16, 301). This is projected to a $d_{\text{model}} = 192$ sequence and fed to a 4-layer pre-norm Transformer encoder (4 heads, FFN dim 768, dropout 0.1). Attentive statistics pooling produces a 384-dimensional embedding, followed by batch normalisation. Total parameters: 2.3M.

Training objective. In addition to CE, one Tier 2 configuration uses **AAM-Softmax** (ArcFace) [7] with angular margin $m = 0.2$ and scale $s = 10$. During training, the target class logit is pushed to $\cos(\theta + m)$; during inference, plain cosine similarity $\times s$ is used for classification.

4.3 Tier 3, Fine-Tuned Pre-Trained Speech Models

XLS-R-300M. We fine-tune XLS-R 300M [3], a 300M-parameter self-supervised model pre-trained on 436 languages. The convolutional feature extractor (4.2M parameters) is frozen; the 24-layer Transformer encoder and a downstream head are trained. The last hidden state is mean-pooled over the time dimension to produce a 1024-dimensional embedding. Two objective variants are evaluated: CE and AAM-Softmax ($m = 0.2$, $s = 10$). Training uses layer-wise learning rates (encoder 10^{-5}, head 10^{-3}), effective batch size 32 via gradient accumulation, mixed precision, and a linear warmup followed by cosine decay. We initialize from the Hugging Face checkpoint `facebook/wav2vec2-xls-r-300m`.

Whisper-small. We fine-tune Whisper-Small [12], an 87M-parameter encoder pre-trained with weakly supervised multilingual ASR on 680k hours of labelled audio. The original positional embedding table (1500 tokens, designed for 30-second windows) is truncated to 150 tokens to match the 3-second input. A two-layer classification head (Linear(768, 256) → ReLU → Dropout(0.3) → Linear(256, 8)) is appended. Training proceeds in two phases: Phase 1 (4 epochs) trains the head only with the encoder frozen; Phase 2 (6 epochs) fine-tunes all parameters with layer-wise learning rates (encoder 10^{-5}, head 10^{-4}) and cosine annealing. We use the `openai/whisper-small` checkpoint.

5 Experimental Setup

All experiments use random seed 42. Log-Mel features are precomputed and stored as float16 NumPy arrays to avoid CPU-decode bottlenecks during training. The XLS-R experiments consume raw 16 kHz waveforms; the Whisper experiments use the Whisper feature extractor to produce (80, 300) log-Mel spectrograms.

Key hyperparameters are summarised in Table 3.

Table 3: Hyperparameter summary across tiers.

	Tier 1	Tier 2	Tier 3
Batch size	32–128	128	8 (accum×4)
Optimizer	AdamW	AdamW	AdamW (layer-wise)
LR	3×10^{-4}	3×10^{-4}	enc 10^{-5}, head 10^{-3}
LR schedule	Cosine	Warmup+Cosine	Warmup+Cosine
Grad clip	–	5.0	5.0
Epochs	20–30	15–40	10
Precision	AMP	AMP	AMP

Experiments were run on NVIDIA Tesla T4 GPUs (16 GB VRAM) via Google Colab and Kaggle Notebooks. Tier 1 and Tier 2 training takes approximately 1.5–4 hours per configuration; Tier 3 XLS-R takes approximately 11 hours for 10 epochs; Tier 3 Whisper takes approximately 1 hour per epoch.

6 Results

6.1 In-Domain Performance (MCV Test Set)

Table 4 reports macro-F1 and micro-accuracy on the in-domain MCV test set for all configurations.

Table 4: In-domain test set results (MCV, 107 439 samples). Macro-F1 is the primary metric. Best result per tier in **bold**.

Tier	Configuration	Macro-F1	Micro-Acc
1	CNN + Log-Mel + CE	0.272	0.405
	CNN + Log-Mel + LS	0.246	0.328
	CNN + MFCC + CE	**0.311**	**0.427**
2	CNN + BiLSTM + CE	0.232	0.372
	CNN + BiLSTM + Mixup + CE	0.238	0.378
	CNN + Transformer + AAM	**0.306**	**0.425**
3	XLS-R-300M + CE	0.480†	0.609
	XLS-R-300M + AAM-Softmax	0.613	0.662
	Whisper-small + CE	**0.795**	**0.848**

† Sub-optimal checkpoint; true peak estimated at 0.55–0.60.

6.2 Out-of-Domain Performance

Table 5 reports OOD results on FLEURS. Note that FLEURS is approximately class-balanced, so macro-F1 figures are not directly comparable to the test set numbers in Table 4.

Table 5: Out-of-domain FLEURS results (6 157 samples). Best in **bold**.

Tier	Configuration	Macro-F1	Micro-Acc
1	CNN + Log-Mel + LS	0.166	0.190
2	CNN + BiLSTM + Mixup + CE	0.249	0.300
	CNN + Transformer + AAM	0.342	0.340
3	XLS-R-300M + AAM-Softmax	0.563	0.568
	Whisper-small + CE	**0.817**	**0.822**

6.3 Per-Language Performance

Table 6 shows per-language F1 on the in-domain test set for the best configuration at each tier.

Table 6: Per-language F1 on the in-domain test set, best configuration per tier. T1: CNN+MFCC+CE. T2: CNN+Transformer+AAM. T3: Whisper-small+CE.

Lang	Tier 1	Tier 2	Tier 3
be (Belarusian)	0.608	0.752	0.878
bg (Bulgarian)	0.127	0.039	0.722
cs (Czech)	0.382	0.550	**0.908**
mk (Macedonian)	0.185	0.114	0.728
pl (Polish)	0.421	0.244	0.888
ru (Russian)	0.406	0.449	0.857
sr (Serbian)	0.123	0.170	0.604
uk (Ukrainian)	0.232	0.126	0.779
Macro-F1	**0.311**	**0.306**	**0.795**

6.4 Confusion Matrices

Figure 1 illustrates the normalized confusion matrices for the Whisper-small model. On the in-domain test set (Figure 1a), performance remains highly diagonal, with primary errors confined to within-branch confusions—particularly the East Slavic cluster. This strong diagonal structure is largely preserved on the out-of-domain FLEURS benchmark (Figure 1b), verifying the model's robust cross-domain generalization.

6.5 Embedding Visualisation

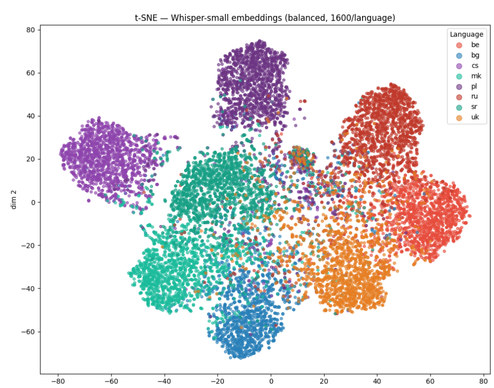
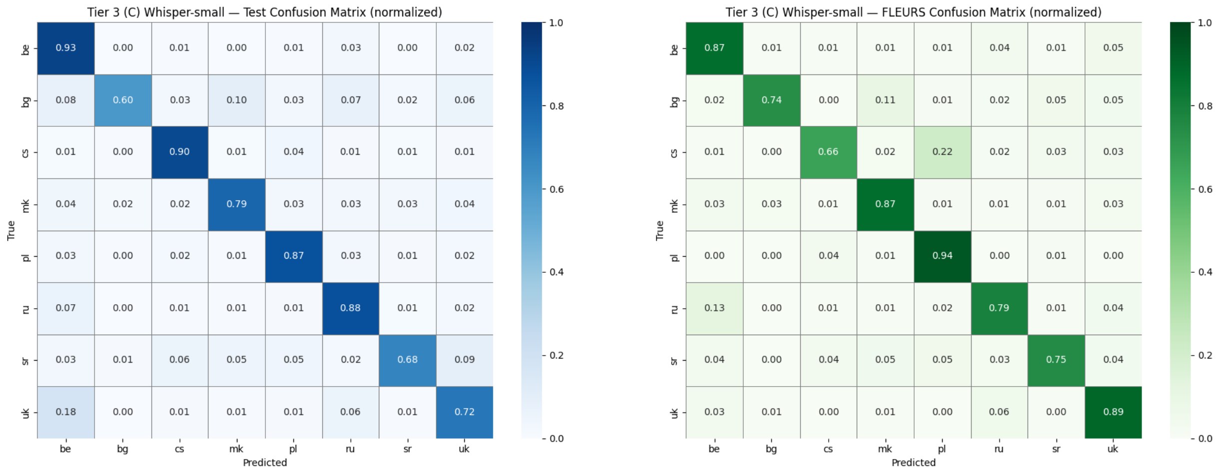


Figure 2: t-SNE of 1000 penultimate-layer embeddings (125 per language) extracted from the Whisper-small model on the test set.

7 Analysis and Discussion

This section analyses results along two orthogonal axes: the effect of each modelling technique on overall performance (Section 7.1), and the contribution of individual language properties to per-class outcomes (Section 7.2).



(a) In-domain test set.

(b) FLEURS set (OOD).

Figure 1: Normalised confusion matrices for the Tier 3 Whisper-small model.

7.1 Technique-Level Analysis

7.1.1 Tier 1: Shallow CNNs and the Feature Representation Bottleneck

All three Tier 1 configurations achieve macro-F1 between 0.246 and 0.311, clustered far below the 80–90% target that a well-functioning LID system should reach for these languages. Train accuracy exceeds 75% in all cases, while test macro-F1 stalls below 0.35, confirming severe overfitting. The bottleneck at this tier is model capacity and the absence of a temporal inductive bias, not the choice of features or objective.

Log-Mel vs. MFCC. The MFCC variant (macro-F1 0.311) outperforms log-Mel (0.272) by $+0.039$. Log-Mel spectrograms represent the full filter-bank energy in each frame as a 2-D image; the CNN must implicitly learn temporal dynamics from the spatial layout of the spectrogram. MFCC features with first-order (Δ) and second-order ($\Delta\Delta$) derivatives encode rate-of-change information explicitly: Δ captures how spectral content moves over time (consonantal transitions, vowel formant trajectories), while $\Delta\Delta$ captures acceleration. For a model without any recurrent or attention component, this explicit temporal summary appears to provide a meaningful advantage. The improvement is modest in absolute terms – both representations fail similarly at this tier – suggesting that feature engineering alone cannot overcome the generalization deficit of a 390k-parameter model on an 8-class problem with 17.5$\times$ test imbalance.

Label smoothing ($\varepsilon = 0.1$)**.** The label-smoothed CNN scores macro-F1 0.246, lower than the plain CE baseline of 0.272. Label smoothing prevents overconfidence by replacing hard one-hot targets with a softer distribution, which in theory improves generalization. However, this benefit only materialises when the model is already close to forming strong, well-separated decision boundaries — a regime this small CNN never reaches. Training was also stopped early at 20 epochs (vs. 30 for the baseline) once it was clear the model was not improving, which makes this a somewhat unclean ablation. The result is best read as: label smoothing does not rescue a capacity-limited model, and may even hurt by making an already-noisy training signal noisier.

SpecAugment. An ablation in the log-Mel configuration finds that SpecAugment provides $+0.008$ macro-F1. The improvement is real but small. SpecAugment masks random frequency bands and time steps, encouraging the model to rely on distributed rather than localised spectral cues. At Tier 1 capacity, the model is unable to exploit fine-grained temporal patterns regardless – the generalisation gap is primarily one of model depth, not data augmentation.

7.1.2 Tier 2: Sequence Modelling and the Architecture–Loss Confound

Tier 2 models incorporate explicit temporal modelling, yet macro-F1 remains between 0.232 and 0.306. The absolute improvement over Tier 1 is minimal, indicating that the bottleneck has shifted but not been eliminated: 2-3M parameter from-scratch models still lack the capacity to learn language-level phonological abstractions from 3-second clips across eight closely related languages.

CNN + BiLSTM. Adding a 2-layer bidirectional LSTM after the CNN frontend (3.0M parameters total) yields macro-F1 0.232, *lower* than the best Tier 1 result of 0.311. Two factors explain this. First, the BiLSTM adds a substantial parameter count that exacerbates overfitting without a corresponding increase in training data; train accuracy exceeds 85% while test macro-F1 stays at 0.23. Second, the BiLSTM processes the full 301-frame sequence (3 seconds at 10 ms hop), and for a from-scratch model this long context may be counterproductive – the hidden states must compress language identity across a highly variable acoustic surface without the prior knowledge that a pre-trained model would bring.

Mixup augmentation. Adding Mixup ($\alpha = 0.2$) to the BiLSTM configuration provides $+0.006$ macro-F1, which is not meaningfully different from zero. Mixup constructs training examples as convex combinations of two randomly sampled clips and their labels. It was originally shown to improve calibration and reduce adversarial sensitivity in image classification, and has since been applied to speech [11]. In this setting, however, the dominant failure mode is systematic class confusion driven by linguistic similarity, not overconfidence or poor calibration. Mixing, for example, a Belarusian clip with a Ukrainian clip produces a synthetic example whose label ($\lambda \cdot \text{be} + (1 - \lambda) \cdot \text{uk}$) does not correspond to any real language and may in fact reinforce the model's tendency to blur the boundary between these two already-confused languages. The null result is therefore unsurprising and consistent with the hypothesis that augmentation cannot substitute for model capacity in this regime.

CNN + Transformer + AAM-Softmax. The Transformer configuration (2.3M parameters) achieves macro-F1 0.306, outperforming the BiLSTM (0.232) with *fewer* parameters. Two modifications are made simultaneously – the architecture (BiLSTM $\rightarrow$ Transformer) and the loss function (CE $\rightarrow$ AAM-Softmax) – which prevents clean attribution, but both changes point in the same direction.

The Transformer's multi-head self-attention computes pairwise interactions across all 301 time steps in $O(T^2)$, allowing it to directly compare any two moments in the 3-second clip regardless of distance. This is a better fit for LID than BiLSTM hidden state propagation, which must carry information about early frames forward through a sequential bottleneck. Language-identifying cues (prosodic patterns, characteristic phoneme sequences, specific vowel systems) are often non-local and sparse across the clip; self-attention can select and aggregate them without the signal degradation that affects long LSTM chains.

AAM-Softmax encourages the model to learn embeddings where each language's cluster is compact and well-separated from the others — directly aligned with the LID objective. Under plain CE, the model is penalised only for wrong predictions, with no direct pressure on the shape of the embedding space. AAM-Softmax adds an angular margin $m = 0.2$: to classify a sample correctly, the model must not just be on the right side of the boundary, but clearly inside it. This pushes same-language embeddings closer together and different-language embeddings further apart. The combination of a richer encoder and a geometry-aware loss moves the Tier 2 best result meaningfully above the BiLSTM baseline (0.306 vs. 0.232), though both remain far from the Tier 3 results.

Attentive statistics pooling. Both Tier 2 architectures use attentive statistics pooling [10] to turn a variable-length sequence into a single fixed-size vector. Instead of averaging all frames equally, a small attention network learns to up-weight the frames that are most useful for identifying the language. The pooling produces the weighted mean and standard deviation of the encoder outputs, where the standard deviation captures how much the features vary over time — a proxy for prosodic variability. This approach is standard in speaker verification [8] and transfers naturally to LID.

7.1.3 Tier 3: The Impact of Pre-Training

The jump from Tier 2 to Tier 3 is far larger than the jump from Tier 1 to Tier 2, which demonstrates that the fundamental limitation of earlier tiers was not architecture or loss design but the absence of language-relevant pre-training knowledge.

XLS-R-300M: self-supervised pre-training. XLS-R-300M is pre-trained on 436 languages using a self-supervised objective: random parts of the audio are masked, and the model learns to predict what was hidden. This produces general acoustic representations that capture detailed phonetic information without needing any language labels. Fine-tuning for LID then requires the model to re-organise those representations around language identity, which needs a fair amount of task-specific signal.

With AAM-Softmax, XLS-R achieves macro-F1 0.613 in-domain and 0.563 OOD – a dramatic improvement over Tier 2 (0.306). The 1024-dimensional mean-pooled embedding provides a rich geometric space for AAM-Softmax to enforce angular margins, and the pre-trained encoder already encodes phonemic contrasts that are relevant to distinguishing Slavic languages. The remaining gap relative to Whisper-small is attributable to the mismatch between XLS-R's self-supervised pre-training objective (predict masked audio) and the downstream task (identify language from acoustics).

Whisper-small: supervised multilingual ASR pre-training. Whisper-small achieves macro-F1 0.795 in-domain and 0.817 OOD, outperforming XLS-R by 0.182 and 0.254 respectively, despite being 3.5× smaller in parameter count. This result is the clearest finding of the study: *the relevance of the pre-training task matters more than model size*.

Whisper is trained with weakly supervised multilingual ASR on 680k hours of labelled speech. The training objective directly requires the encoder to represent what language is being spoken (the decoder conditions on a language token), what phonemes are being uttered (the ASR transcription), and how prosody and timing encode meaning. All of these dimensions are relevant for LID. By the time Whisper is fine-tuned for classification, its encoder already partitions the acoustic space in a way that respects language boundaries — the classification head merely needs to learn a mapping from this pre-organised space to class labels.

The two-phase fine-tuning strategy (4 epochs head-only, then 6 epochs full fine-tuning) is also likely important. Phase 1 initializes the classification head to a reasonable starting point before the encoder weights are updated, preventing large gradients from the randomly initialized head from disrupting the pre-trained encoder representations early in training. This approach mirrors standard practice in transfer learning for vision and NLP.

Pre-Training Task Alignment: a general principle. Taken together, the Tier 3 results support the following hierarchy of pre-training benefit for Slavic LID: *supervised multilingual ASR on relevant languages > self-supervised pre-training on many languages > from-scratch training*. This has a direct practical implication: when selecting a backbone for low-resource or related-language LID tasks, practitioners should prioritise coverage of target languages in the pre-training data over raw model capacity.

7.2 Language-Level Analysis

The Slavic language family provides a natural three-level hierarchy for error analysis: within-branch confusion (most common), cross-branch confusion between adjacent branches (occasional), and cross-branch confusion between distant branches (rare). We analyse each branch in turn before examining cross-branch patterns and OOD behaviour.

7.2.1 East Slavic Branch: Belarusian, Russian, Ukrainian

The persistent confusion cluster. Belarusian, Russian, and Ukrainian are the three most similar languages in the study. They share the a common lexical core derived from Old Church Slavonic and Proto-Slavic, and closely related phonological inventories including the same set of consonant palatalisation contrasts and similar vowel systems. In terms of acoustic surface, they differ primarily in rhythm, vowel reduction patterns (Russian reduces unstressed vowels more aggressively than Belarusian or Ukrainian), and specific phoneme realisations (the Ukrainian */g/* realised as a fricative *[h]*, which contrasts with Russian's plosive *[g]*).

These differences are real but subtle in 3-second clips sampled from crowdsourced speech, where speaker variability, microphone quality, and reading style all introduce variation that may exceed the between-language signal. At Tier 3 B (XLS-R + AAM-Softmax), 14.7% of Ukrainian samples are misclassified as Belarusian and 12.7% as Russian – a 27.4% combined error rate within the East Slavic cluster. Russian achieves the highest within-branch F1 (0.741 at Tier 3 B), likely because it is the most represented language in international speech datasets used for XLS-R pre-training, giving the model a stronger acoustic prior for Russian phonology.

Belarusian: high recall, impure predictions. Belarusian is the largest class in the test set (29,232 samples) and consistently achieves high recall across all models, but at the cost of absorbing many Ukrainian and Russian clips. At Tier 3 B, Belarusian precision is 0.835 while recall is 0.689, reflecting a model that correctly identifies most Belarusian clips but also incorrectly labels many Ukrainian clips as Belarusian. This pattern is consistent with the class imbalance: the model learns that predicting Belarusian is frequently rewarded, creating an implicit prior toward that label for ambiguous East Slavic clips.

Ukrainian: squeezed between neighbours. Ukrainian is systematically under-predicted at all tiers. Even at Tier 3 C (Whisper-small), Ukrainian achieves F1=0.779, the second-lowest score among the eight languages for that model. Ukrainian is phonologically intermediate between Belarusian and Russian: its vowel inventory is closer to Belarusian, while its consonantal system shares features with Russian. This means Ukrainian clips are plausible candidates for both neighbours in the East Slavic cluster, and without strong prosodic or lexical cues in the 3-second window, classifiers place Ukrainian samples in the wrong bucket.

7.2.2 West Slavic Branch: Czech and Polish

Polish: distinctiveness as a robustness advantage. Polish is the easiest West Slavic language to identify across all models and all conditions (in-domain and OOD). At Tier 3 C, Polish achieves F1=0.888 in-domain and 0.834 on FLEURS – the best or near-best score in both splits. This robustness is directly attributable to Polish phonology: Polish has retained nasal vowels (/õ/ and /ẽ/), which are absent in all other Slavic languages studied here, and has an exceptionally dense consonant cluster system (sequences of 4–6 consonants occur in common words). These features produce a highly distinctive acoustic signature in spectrograms and waveforms. Nasal vowels create a characteristic spectral resonance pattern, and consonant clusters generate burst sequences that are visually and acoustically distinctive. Even Tier 1 models, which cannot learn language-level abstractions, learn to respond to these low-level spectral patterns, giving Polish a consistently strong F1 at every tier (0.421 at Tier 1, rising to 0.888 at Tier 3 C despite the much weaker model at Tier 1). The stability of Polish recognition across recording conditions (MCV crowdsourced vs. FLEURS studio) confirms that the identifying features are phonological rather than recording-condition-specific.

Czech: high in-domain performance, catastrophic OOD collapse. Czech achieves the highest per-language F1 among all languages in the study under Whisper-small in-domain (0.908), yet collapses to 0.744 on FLEURS — the largest OOD drop of any language for the Whisper model. Table 7 shows this pattern is consistent across models.

Table 7: Czech (cs) F1 on in-domain test vs. FLEURS (OOD).

Model	Test F1	FLEURS F1
CNN + Transformer + AAM (Tier 2)	0.550	0.123
XLS-R + AAM-Softmax (Tier 3)	0.684	0.203
Whisper-small (Tier 3)	**0.908**	**0.744**

Two factors could contribute to this pattern. First, the Czech MCV corpus used for training and in-domain testing predominantly contains a specific recording demographic – native Czech speakers reading prepared sentences into a consumer-grade microphone – creating a strong speaker-style prior. The FLEURS Czech recordings use a different read-speech style with different studio acoustics, and models over-fitted to the MCV acoustic profile fail to transfer. Second, when Czech is misclassified on FLEURS under the weaker Tier 2 model, it is predominantly predicted as Serbian – a cross-branch error that never occurs on the in-domain test. This suggests that in the FLEURS recording condition, Czech loses the phonological cues that distinguish it from South Slavic languages (primarily its front vowel quality and specific sibilant system), possibly because studio conditions reduce the high-frequency spectral detail on which the model relies. Whisper-small's superior OOD Czech performance (0.744 vs. 0.223) indicates that its ASR pre-training encodes more robust phonemic representations that partially survive the domain shift.

7.2.3 South Slavic Branch: Bulgarian, Macedonian, Serbian

Structural disadvantage from test imbalance. All three South Slavic languages are minority classes in the test set: Bulgarian (6,318), Macedonian (3,263), and Serbian (1,666) together constitute only 10.4% of test samples, compared to the three East Slavic languages which account for 60.2%. This imbalance systematically disadvantages South Slavic languages at every tier via two mechanisms. First, the model receives fewer gradient updates from South Slavic examples during training, even though the training set is balanced — the test distribution does not reflect training experience, so the model learns to be uncertain about South Slavic instances. Second, at test time, a model that is uncertain between a South Slavic language and a large East Slavic class will preferentially predict the East Slavic class simply because the implied prior probability is higher under the implicit label-frequency weighting of cross-entropy loss.

Bulgarian: recovery at Tier 3. Bulgarian achieves the most dramatic improvement across tiers: F1=0.127 at Tier 1, 0.066 at Tier 2 (worse than Tier 1, reflecting the severity of the imbalance problem at that capacity), and 0.722 at Tier 3 C. Bulgarian is a South Slavic language with a specific phonological feature that distinguishes it from East Slavic: it has lost grammatical case endings in favour of analytic constructions (like English), and its vowel system includes a distinctive mid-central vowel /ə/ that does not appear in East Slavic languages. However, these distinctions are subtle in 3-second clips from crowd-sourced speech. The Tier 1 and Tier 2 models, lacking pre-trained language knowledge, cannot reliably identify these cues and default to predicting more common classes. Whisper-small's ASR pre-training, which includes Bulgarian, gives it explicit knowledge of Bulgarian phonemes and allows it to correctly identify the language even from short clips.

Macedonian: the most confused language at Tier 2. Macedonian achieves the lowest F1 at Tier 2 (0.114 under the best Tier 2 model). Macedonian is closely related to both Bulgarian (they share the loss of case morphology and many lexical items) and Serbian (geographic proximity, mutual intelligibility). The model at Tier 2 has no principled basis for distinguishing between these languages from acoustic properties alone — the differences are primarily lexical and in fine prosodic details. Macedonian also has the second-smallest test set (3,263 samples), compounding the imbalance effect. At Tier 3 C (Whisper-small), Macedonian reaches F1=0.728, indicating that ASR pre-training provides sufficient phonemic and prosodic knowledge to distinguish Macedonian from its nearest neighbours.

Serbian: persistent underperformance. Serbian is the hardest language in the study across all tiers. Even under Whisper-small, Serbian achieves F1=0.604 — the lowest score of any language at Tier 3. Serbian is the smallest class in the test set (1,666 samples), which directly limits the model's exposure to Serbian-specific acoustic patterns. Beyond imbalance, Serbian shares a close relationship with Croatian and Bosnian (which are absent from the dataset and may appear in pre-training corpora as related languages), potentially confusing pre-trained representations. MCV Serbian recordings may come from speakers of different dialect groups (Štokavian, Chakavian, Kajkavian), introducing acoustic variability that is not present for most other languages in this study.

7.2.4 Cross-Branch Confusion Patterns

Under lower-tier models, cross-branch confusions are common and reveal an important linguistic structure. The dominant cross-branch errors at Tier 2 are:

- **Bulgarian → Russian/Belarusian**: Bulgarian uses Cyrillic, shares some phonological features with Russian, and has a similar vowel-reduction pattern. Without fine-grained discriminative capacity, models conflate these.
- **Czech → Serbian** (on FLEURS): A surprising cross-branch confusion discussed in Section 7.2. Czech sibilants and Serbian sibilants overlap acoustically under certain recording conditions.
- **Macedonian → Bulgarian/Serbian**: Geographic and structural proximity makes Macedonian indistinguishable from its neighbours for low-capacity models.

At Tier 3 C (Whisper-small), cross-branch confusions are substantially reduced: the confusion matrix (Figure 1a) is predominantly diagonal with off-diagonal mass concentrated within branches rather than across them.

7.2.5 The Effect of the 3-Second Clip Constraint

All models are trained and evaluated on fixed 3-second clips. This constraint affects languages differently. Languages with short characteristic phonemic markers (Polish nasal vowels, Czech front vowels) can be identified reliably within 3 seconds. Languages whose identity is most clearly expressed through prosodic rhythm, lexical choice, or morphological patterns (e.g., Macedonian vs. Bulgarian) require more context and are harder to identify from a short window. The 3-second constraint is therefore not language-neutral: it systematically disadvantages languages whose discriminative cues operate at longer timescales.

To quantify this, we ran a clip-length ablation using the Whisper-small model (Table 8). Each test clip was either zero-padded to 3 s ("padded") or evaluated with only the available frames visible ("masked"). Performance drops steeply below 2 seconds: a 1-second clip yields macro-F1 of only 0.400 under masking, less than half the full-clip score. Even at 2 seconds the model achieves 0.701, confirming that the full 3-second window is important. Zero-padding is consistently worse than masking (e.g. -0.047 at 2 s), since the model interprets the silence padding as a speech signal and is misled by it.

Table 8: Whisper-small macro-F1 at different clip lengths. "Padded" zero-pads shorter clips to 3 s; "masked" uses only the available frames.

Length (s)	Padded	Masked
0.5	0.175	0.236
1.0	0.325	0.400
2.0	0.654	0.701
3.0	0.795	0.795

7.2.6 Branch-Level Summary

Table 9 summarises average per-branch F1 at each tier for the best configuration.

Table 9

Branch	Tier 1	Tier 2	Tier 3
East (be, ru, uk)	0.415	0.442	0.838
West (cs, pl)	0.402	0.397	0.898
South (bg, mk, sr)	0.145	0.108	0.685

The South Slavic branch is consistently the hardest, lagging 0.153 F1 points behind East Slavic and 0.213 behind West Slavic at Tier 3. This gap is driven by test set imbalance (all three South Slavic languages are minority classes), linguistic similarity within the branch (especially Macedonian/Bulgarian/Serbian), and the absence of strongly discriminative short-timescale phonological markers of the kind that make Polish or Czech easily identifiable.

8 Bulgarian Dialect Classification

Having established that the fine-tuned Whisper backbone reliably identifies Bulgarian as a language (F1=0.722–

0.812 across evaluation sets), we ask a harder follow-up question: can the same backbone distinguish Bulgarian *dialects* from each other? Dialects present a fundamentally more difficult problem than language identification – the acoustic differences are subtler, labelled training data is far smaller, and there is no sharp boundary between neighbouring dialects.

8.1 Data: BDLT

We use recordings from the *Bulgarian Dialectology as Living Tradition* (BDLT) academic corpus [1], which archives field interviews with native speakers from villages across Bulgaria. We mapped 78 villages to four geographic regions based on the standard Bulgarian dialectological classification:

- **NW** (Northwestern): Western-type dialects where the historical vowel *yat* is always realised as */e/*; characteristic consonant features distinct from standard Bulgarian.
- **NE** (Northeastern): Eastern-type dialects with the standard yat alternation – */ya/* before back syllables, */e/* before front syllables; the basis of literary Bulgarian.
- **SW** (Southwestern): Western-type dialects from the Rhodopes and Sofia region; internally diverse, covering several sub-dialect groups.
- **SE** (Southeastern): Eastern-type dialects from the Thrace and Burgas region; similar yat alternation to NE but with distinct vowel and stress patterns.

After filtering recordings that were too short (fewer than 5 clips), we extracted 5,178 non-overlapping 10-second clips from 197 recordings, split 70/15/15 into train/val/test with speaker-disjoint partitions. The SE class is overrepresented (test: NW=182, NE=198, SW=195, SE=383), so we apply inverse-frequency class weights during training to compensate.

8.2 Method

We re-use the fine-tuned Whisper-small backbone from Tier 3 as a starting point, replacing the 8-way language head with a 4-way dialect head (Linear(768, 256) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear(256, 4)). The same two-phase training is applied: Phase 1 (4 epochs) trains only the head with the encoder frozen, then Phase 2 (6 epochs) fine-tunes all parameters with layer-wise learning rates (encoder 10^{-5}, head 10^{-4}). Clips are 10 seconds long (1,000 Mel frames).

8.3 Results

The model reaches macro-F1 **0.867** and accuracy **86.95%** on the test set, far above the 25% random baseline (Table 10). Validation accuracy at the best checkpoint was 88.41%. Figure 3 shows the training trajectory: after the encoder is unfrozen at epoch 5, accuracy jumps rapidly from 63% to above 87%, confirming that the backbone representations need only light adaptation for the dialect task.

Table 10: Bulgarian 4-way regional dialect classification on the BDLT test set.

Region	Test F1	Support
NW (Northwestern)	0.888	182
NE (Northeastern)	0.873	198
SW (Southwestern)	0.827	195
SE (Southeastern)	0.880	383
Macro-F1	**0.867**	958
Accuracy	**0.870**	

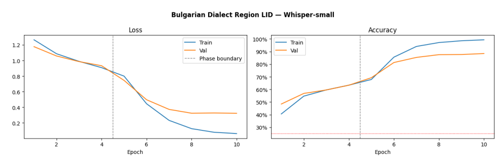


Figure 3: Training and validation curves for the Bulgarian dialect model. The vertical dashed line marks the start of full fine-tuning (Phase 2).

8.4 Discussion

SW is the hardest region. Southwestern Bulgarian (F1=0.827) is the most often confused dialect. This is linguistically expected: the SW region encompasses a wide variety of sub-dialects (Rhodopean, Transitional, Sofia-area), making the region label internally heterogeneous. The model is asked to lump together speakers whose dialects are acoustically quite different from each other.

NW and NE are most distinct. Northwestern (F1=0.888) and Northeastern (F1=0.873) are the best-classified regions. These are the two poles of the main East–West dialectal split in Bulgarian: the yat reflex is starkly different between them, providing a strong acoustic signal that the model can reliably pick up from a 10-second clip.

Confusion pattern follows geography. Figure 4 shows that most errors stay within geographically adjacent regions (NW$\leftrightarrow$SW, NE$\leftrightarrow$SE), consistent with the gradual nature of dialect continua. True cross-axis confusions (NW$\leftrightarrow$NE or SW$\leftrightarrow$SE) are rare.

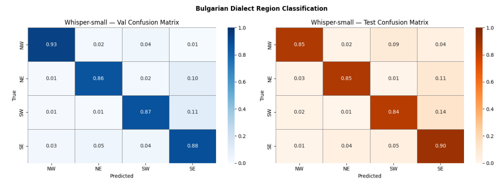


Figure 4: Normalised confusion matrices for the Bulgarian dialect model (left: validation set, right: test set).

Embedding structure. The t-SNE plot of test embeddings (Figure 5) shows four well-separated clusters, with the expected proximity between the two Western dialects (NW, SW) and the two Eastern dialects (NE, SE) – matching the fundamental East–West axis of Bulgarian dialectology.

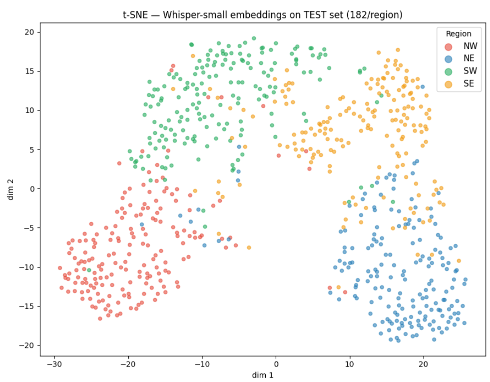


Figure 5: t-SNE of dialect model embeddings on the test set (182 samples/region). Western dialects (NW, SW) and Eastern dialects (NE, SE) cluster on opposite sides.

9 Conclusion

We evaluated nine deep learning configurations for 8-way Slavic language identification across three model tiers. The best system, a fine-tuned Whisper-small encoder, achieves macro-F1 of 0.795 in-domain and 0.817 out-of-domain, demonstrating that supervised multilingual pre-training on a task-relevant distribution is the single most important design choice – more impactful than model size, loss function, or input representation. A clip-length ablation confirms that 3-second windows are essential: macro-F1 falls to 0.40 at 1 second and 0.24 at 0.5 seconds.

From-scratch models (Tiers 1–2) are severely capacity-limited: all configurations cluster between macro-F1 0.23 and 0.31 regardless of architecture or objective. The persistent East Slavic confusion cluster (be/ru/uk) and Czech out-of-domain collapse highlight language-specific challenges that warrant future work, particularly on domain-robust training and class-balanced evaluation.

As a supplementary experiment, we applied the same Whisper backbone to 4-way Bulgarian dialect classification using the BDLT corpus, achieving macro-F1 0.867 with only $\sim$200 source recordings. This suggests that task-aligned pre-training generalises well beyond language-level distinctions and may be useful for dialect-level applications.

Future directions include: (1) applying Whisper-medium or Whisper-large for higher-capacity fine-tuning; (2) using class-weighted sampling or class-aware AAM-Softmax to mitigate the 17.5$\times$ test imbalance; and (3) investigating the influence of recording channel and speaker demographics on cross-domain performance.

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References

[1] Institute for Bulgarian Language, Bulgarian Academy of Sciences, "Bulgarian Dialectology as Living Tradition (BDLT)," http://bdlt.nbu.bg, Sofia University St. Kliment Ohridski, accessed 2026.

[2] R. Ardila et al., "Common Voice: A massively multilingual speech corpus," *Proc. LREC*, 2020.

[3] A. Babu et al., "XLS-R: Self-supervised cross-lingual speech representation learning at scale," *Proc. Interspeech*, 2022.

[4] A. Baevski et al., "wav2vec 2.0: A framework for self-supervised learning of speech representations," *Proc. NeurIPS*, 2020.

[5] C. Bartz et al., "Language identification using deep convolutional recurrent neural networks," *ICONIP*, 2017.

[6] A. Conneau et al., "FLEURS: Few-shot learning evaluation of universal representations of speech," *Proc. SLT*, 2022.

[7] J. Deng et al., "ArcFace: Additive angular margin loss for deep face recognition," *Proc. CVPR*, 2019.

[8] B. Desplanques et al., "ECAPA-TDNN: Emphasized channel attention, propagation and aggregation in TDNN based speaker verification," *Proc. Interspeech*, 2020.

[9] H. Li et al., "Spoken language recognition: From fundamentals to practice," *Proc. IEEE*, 101(5):1spoken–1380, 2013.

[10] K. Okabe et al., "Attentive statistics pooling for deep speaker embedding," *Proc. Interspeech*, 2018.

[11] D. S. Park et al., "SpecAugment: A simple data augmentation method for automatic speech recognition," *Proc. Interspeech*, 2019.

[12] A. Radford et al., "Robust speech recognition via large-scale weak supervision," *Proc. ICML*, 2023.

[13] S. Revay and M. Teschke, "Multiclass language identification using deep learning on spectral images of audio signals," *arXiv:1905.04348*, 2019.

[14] D. Snyder et al., "X-vectors: Robust DNN embeddings for speaker recognition," *Proc. ICASSP*, 2018.

[15] J. Valk and T. Alumäe, "VoxLingua107: A dataset for spoken language recognition," *Proc. SLT*, 2021.